

GOVERNMENT ENGINEERING COLLEGE MADHUBANI

ELECTRICAL DEPARTMENT

SUBJECT - ELECTRICAL & ELECTRONICS MEASUREMENT

MODULE 2 NOTES

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AC BRIDGES

2.1 General form of A.C. bridge

AC bridge are similar to D.C. bridge in topology (way of connecting). It consists of four arm AB, BC, CD and DA. Generally the impedance to be measured is connected between 'A' and 'B'. A detector is connected between 'B' and 'D'. The detector is used as null deflection instrument. Some of the arms are variable element. By varying these elements, the potential values at 'B' and 'D' can be made equal. This is called balancing of the bridge.

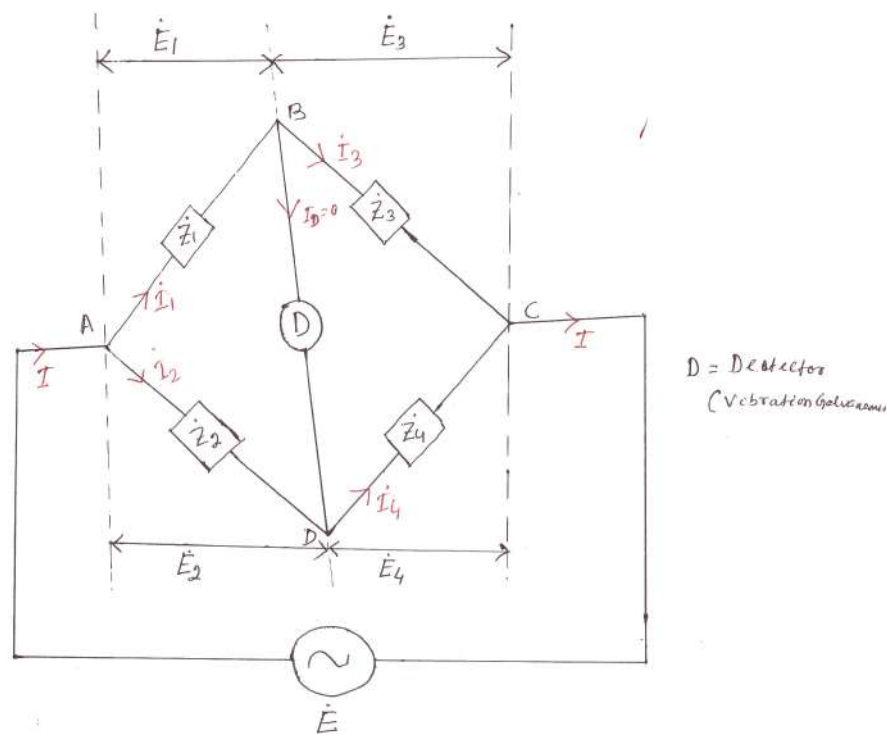


Fig. 2.1 General form of A.C. bridge

At the balance condition, the current through detector is zero.

$$\therefore \dot{I}_1 = \dot{I}_3$$

$$\dot{I}_2 = \dot{I}_4$$

$$\therefore \frac{\dot{I}_1}{\dot{I}_2} = \frac{\dot{I}_3}{\dot{I}_4} \quad (2.1)$$

At balance condition,

Voltage drop across 'AB'=voltage drop across 'AD'.

$$\dot{E}_1 = \dot{E}_2$$

$$\therefore \dot{I}_1 \dot{Z}_1 = \dot{I}_2 \dot{Z}_2 \quad (2.2)$$

Similarly, Voltage drop across 'BC'=voltage drop across 'DC'

$$\dot{E}_3 = \dot{E}_4$$

$$\therefore \dot{I}_3 \dot{Z}_3 = \dot{I}_4 \dot{Z}_4 \quad (2.3)$$

From Eqn. (2.2), we have $\therefore \frac{\dot{I}_1}{\dot{I}_2} = \frac{\dot{Z}_2}{\dot{Z}_1}$ (2.4)

From Eqn. (2.3), we have $\therefore \frac{\dot{I}_3}{\dot{I}_4} = \frac{\dot{Z}_4}{\dot{Z}_3}$ (2.5)

From equation -2.1, it can be seen that, equation -2.4 and equation-2.5 are equal.

$$\therefore \frac{\dot{Z}_2}{\dot{Z}_1} = \frac{\dot{Z}_4}{\dot{Z}_3}$$

$$\therefore \dot{Z}_1 \dot{Z}_4 = \dot{Z}_2 \dot{Z}_3$$

Products of impedances of opposite arms are equal.

$$\therefore |Z_1| \angle \theta_1 |Z_4| \angle \theta_4 = |Z_2| \angle \theta_2 |Z_3| \angle \theta_3$$

$$\Rightarrow |Z_1| |Z_4| \angle \theta_1 + \theta_4 = |Z_2| |Z_3| \angle \theta_2 + \theta_3$$

$$|Z_1| |Z_4| = |Z_2| |Z_3|$$

$$\theta_1 + \theta_4 = \theta_2 + \theta_3$$

- * For balance condition, magnitude on either side must be equal.
- * Angle on either side must be equal.

Summary

For balance condition,

- $\dot{I}_1 = \dot{I}_3, \dot{I}_2 = \dot{I}_4$
- $|Z_1||Z_4| = |Z_2||Z_3|$
- $\theta_1 + \theta_4 = \theta_2 + \theta_3$
- $\dot{E}_1 = \dot{E}_2 \quad \& \quad \dot{E}_3 = \dot{E}_4$

2.2 Types of detector

The following types of instruments are used as detector in A.C. bridge.

- Vibration galvanometer
- Head phones (speaker)
- Tuned amplifier

2.2.1 Vibration galvanometer

Between the point 'B' and 'D' a vibration galvanometer is connected to indicate the bridge balance condition. This A.C. galvanometer which works on the principle of resonance. The A.C. galvanometer shows a dot, if the bridge is unbalanced.

2.2.2 Head phones

Two speakers are connected in parallel in this system. If the bridge is unbalanced, the speaker produced more sound energy. If the bridge is balanced, the speaker do not produced any sound energy.

2.2.3 Tuned amplifier

If the bridge is unbalanced the output of tuned amplifier is high. If the bridge is balanced, output of amplifier is zero.

2.3 Measurements of inductance

2.3.1 Maxwell's inductance bridge

The choke for which R_1 and L_1 have to measure connected between the points 'A' and 'B'. In this method the unknown inductance is measured by comparing it with the standard inductance.

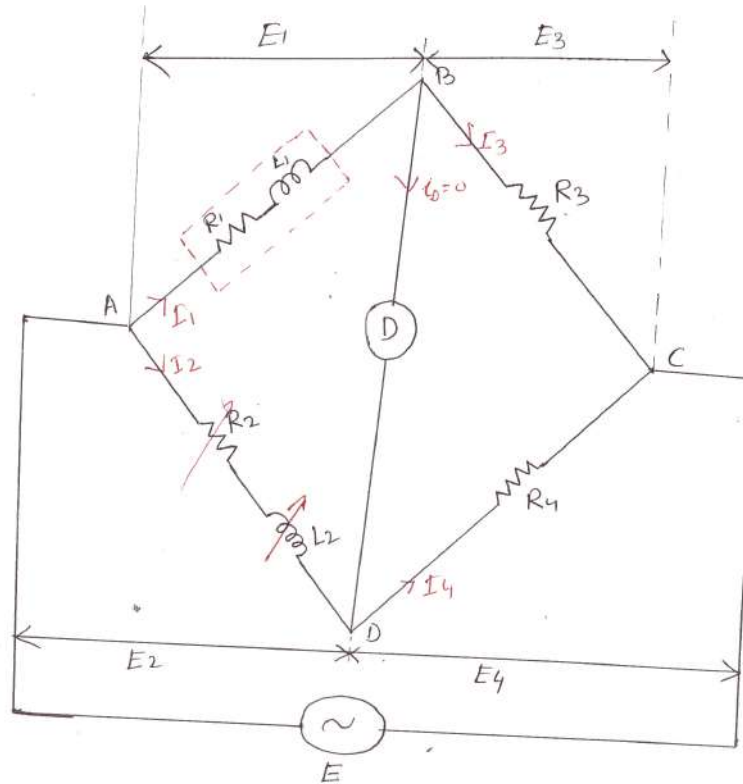


Fig. 2.2 Maxwell's inductance bridge

L_2 is adjusted, until the detector indicates zero current.

Let R_1 = unknown resistance

L_1 = unknown inductance of the choke.

L_2 = known standard inductance

R_1, R_2, R_4 = known resistances.

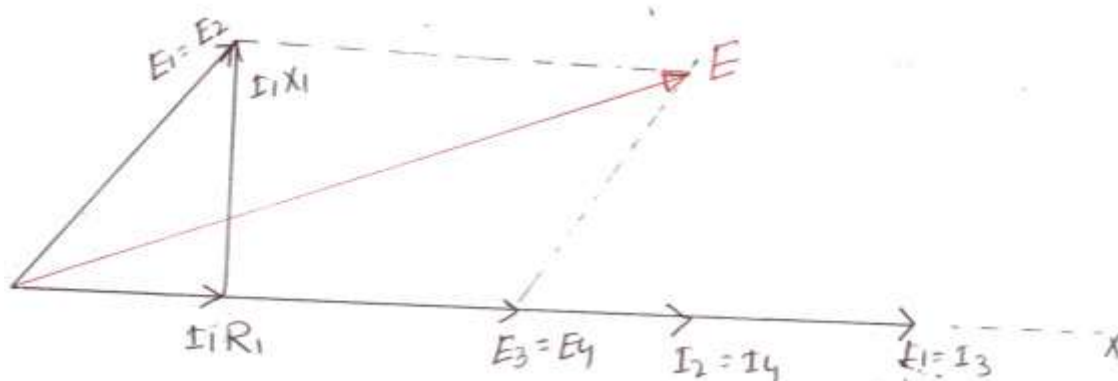


Fig 2.3 Phasor diagram of Maxwell's inductance bridge

At balance condition, $\dot{Z}_1 \dot{Z}_4 = \dot{Z}_2 \dot{Z}_3$

$$(R_1 + jXL_1)R_4 = (R_2 + jXL_2)R_3$$

$$(R_1 + j\omega L_1)R_4 = (R_2 + j\omega L_2)R_3$$

$$R_1R_4 + j\omega L_1R_4 = R_2R_3 + j\omega L_2R_3$$

Comparing real part,

$$R_1R_4 = R_2R_3$$

$$\therefore R_1 = \frac{R_2R_3}{R_4} \quad (2.6)$$

Comparing the imaginary parts,

$$\omega L_1R_4 = \omega L_2R_3$$

$$L_1 = \frac{L_2R_3}{R_4} \quad (2.7)$$

$$\text{Q-factor of choke, } Q = \frac{\omega L_1}{R_1} = \frac{\omega L_2R_3R_4}{R_4R_2R_3}$$

$$Q = \frac{\omega L_2}{R_2} \quad (2.8)$$

Advantages

- ✓ Expression for R_1 and L_1 are simple.
- ✓ Equations are simple
- ✓ They do not depend on the frequency (as ω is cancelled)
- ✓ R_1 and L_1 are independent of each other.

Disadvantages

- ✓ Variable inductor is costly.
- ✓ Variable inductor is bulky.

2.3.2 Maxwell's inductance capacitance bridge

Unknown inductance is measured by comparing it with standard capacitance. In this bridge, balance condition is achieved by varying ' C_4 '.

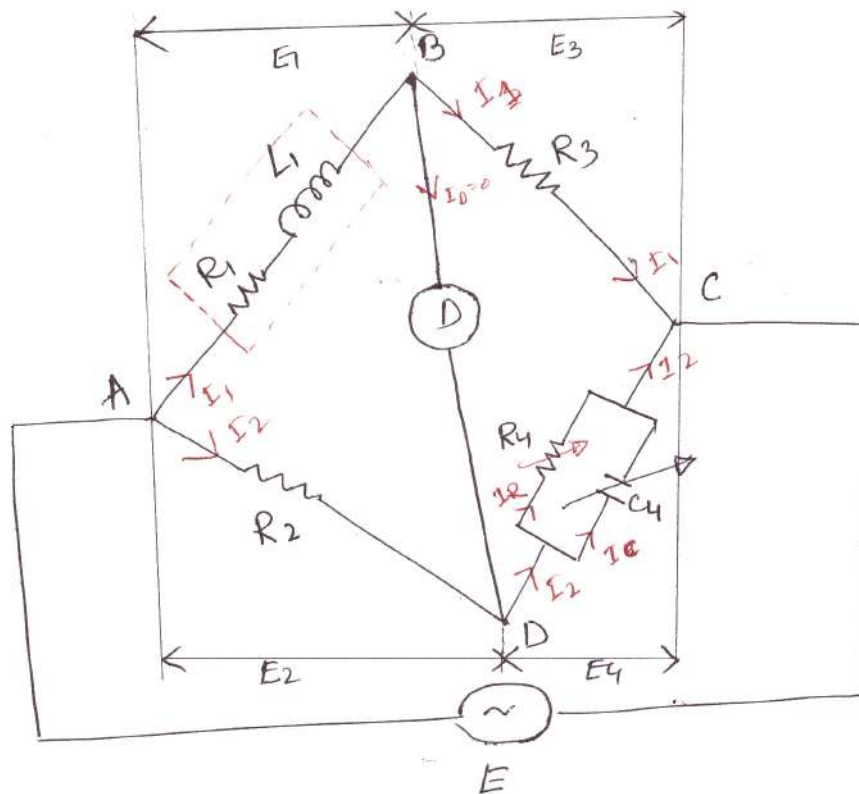


Fig 2.4 Maxwell's inductance capacitance bridge

At balance condition, $Z_1 Z_4 = Z_3 Z_2$ (2.9)

$$Z_4 = R_4 \parallel \frac{1}{j\omega C_4} = \frac{R_4 \times \frac{1}{j\omega C_4}}{R_4 + \frac{1}{j\omega C_4}}$$

$$Z_4 = \frac{R_4}{j\omega R_4 C_4 + 1} = \frac{R_4}{1 + j\omega R_4 C_4} \quad (2.10)$$

∴ Substituting the value of Z_4 from eqn. (2.10) in eqn. (2.9) we get

$$(R_1 + j\omega L_1) \times \frac{R_4}{1 + j\omega R_4 C_4} = R_2 R_3$$

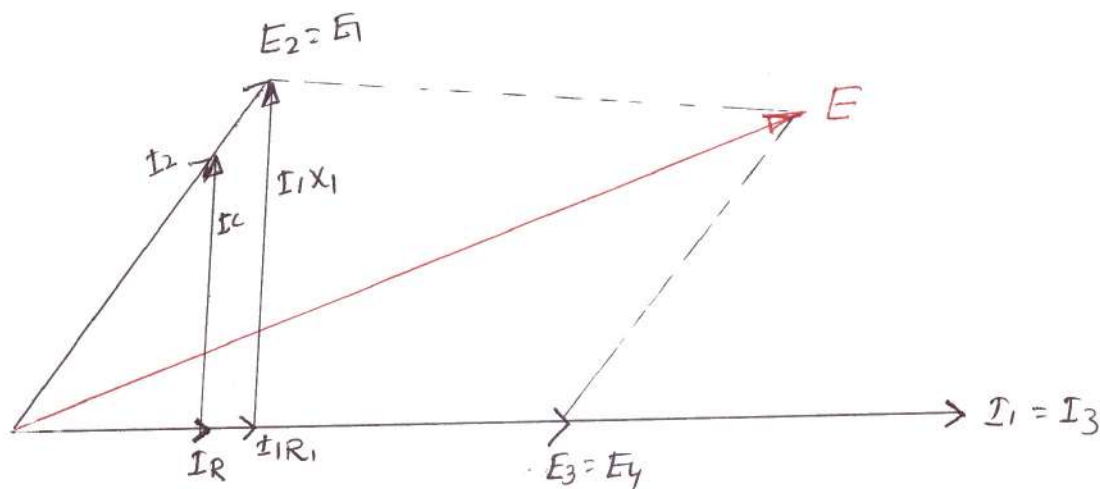


Fig 2.5 Phasor diagram of Maxwell's inductance capacitance bridge

$$(R_1 + j\omega L_1) R_4 = R_2 R_3 (1 + j\omega R_4 C_4)$$

$$R_1 R_4 + j\omega L_1 R_4 = R_2 R_3 + j\omega C_4 R_4 R_2 R_3$$

Comparing real parts,

$$R_1 R_4 = R_2 R_3$$

$$\Rightarrow R_1 = \frac{R_2 R_3}{R_4} \quad (2.11)$$

Comparing imaginary part,

$$\omega L_1 R_4 = \omega C_4 R_4 R_2 R_3$$

$$L_1 = C_4 R_2 R_3 \quad (2.12)$$

Q-factor of choke,

$$Q = \frac{\omega L_1}{R_1} = \omega \times C_4 R_2 R_3 \times \frac{R_4}{R_2 R_3}$$

$$Q = \omega C_4 R_4 \quad (2.13)$$

Advantages

- ✓ Equation of L_1 and R_1 are simple.
- ✓ They are independent of frequency.
- ✓ They are independent of each other.
- ✓ Standard capacitor is much smaller in size than standard inductor.

Disadvantages

- ✓ Standard variable capacitance is costly.
- ✓ It can be used for measurements of Q-factor in the ranges of 1 to 10.
- ✓ It cannot be used for measurements of choke with Q-factors more than 10.

We know that $Q = \omega C_4 R_4$

For measuring chokes with higher value of Q-factor, the value of C_4 and R_4 should be higher. Higher values of standard resistance are very expensive. Therefore this bridge cannot be used for higher value of Q-factor measurements.

2.3.3 Hay's bridge

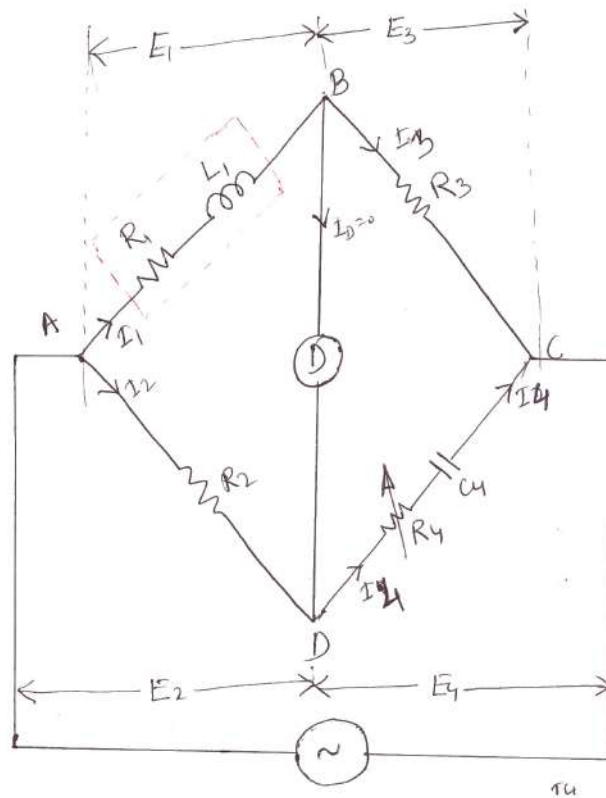


Fig 2.6 Hay's bridge

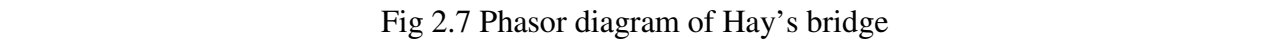
$$\dot{E}_1 = I_1 R_1 + j I_1 X_1$$

$$\dot{E} = \dot{E}_1 + \dot{E}_3$$

$$\dot{E}_4 = I_4 R_4 + \frac{I_4}{j \omega C_4}$$

$$\dot{E}_3 = I_3 R_3$$

$$Z_4 = R_4 + \frac{1}{j \omega C_4} = \frac{1 + j \omega R_4 C_4}{j \omega C_4}$$



At balance condition, $Z_1 Z_4 = Z_3 Z_2$

$$(R_1 + j\omega L_1) \left(\frac{1 + j\omega R_4 C_4}{j\omega C_4} \right) = R_2 R_3$$

$$(R_1 + j\omega L_1)(1 + j\omega R_4 C_4) = j\omega R_2 C_4 R_3$$

$$R_1 + j\omega C_4 R_4 R_1 + j\omega L_1 + j^2 \omega^2 L_1 C_4 R_4 = j\omega C_4 R_2 R_3$$

$$(R_1 - w^2 L_1 C_4 R_4) + j(w C_4 R_4 R_1 + w L_1) = jw C_4 R_2 R_3$$

Comparing the real term,

$$R_1 - w^2 L_1 C_4 R_4 = 0$$

$$R_1 = w^2 L_1 C_4 R_4 \quad (2.14)$$

$$R_1 = w^2 L_1 C_4 R_4 \quad (2.14)$$

Comparing the imaginary terms,

$$wC_4R_4R_1 + wL_1 = wC_4R_2R_3$$

$$C_4R_4R_1 + L_1 = C_4R_2R_3$$

$$L_1 = C_4R_2R_3 - C_4R_4R_1 \quad (2.15)$$

Substituting the value of R_1 from eqn. 2.14 into eqn. 2.15, we have,

$$L_1 = C_4R_2R_3 - C_4R_4 \times w^2L_1C_4R_4$$

$$L_1 = C_4R_2R_3 - w^2L_1C_4^2R_4^2$$

$$L_1(1 + w^2L_1C_4^2R_4^2) = C_4R_2R_3$$

$$L_1 = \frac{C_4R_2R_3}{1 + w^2L_1C_4^2R_4^2} \quad (2.16)$$

Substituting the value of L_1 in eqn. 2.14, we have

$$R_1 = \frac{w^2C_4^2R_2R_3R_4}{1 + w^2C_4^2R_4^2} \quad (2.17)$$

$$Q = \frac{wL_1}{R_1} = \frac{w \times C_4R_2R_3}{1 + w^2C_4^2R_4^2} \times \frac{1 + w^2C_4^2R_4^2}{w^2C_4^2R_4R_2R_3}$$

$$Q = \frac{1}{wC_4R_4} \quad (2.18)$$

Advantages

- ✓ Fixed capacitor is cheaper than variable capacitor.
- ✓ This bridge is best suitable for measuring high value of Q-factor.

Disadvantages

- ✓ Equations of L_1 and R_1 are complicated.
- ✓ Measurements of R_1 and L_1 require the value of frequency.
- ✓ This bridge cannot be used for measuring low Q- factor.

2.3.4 Owen's bridge

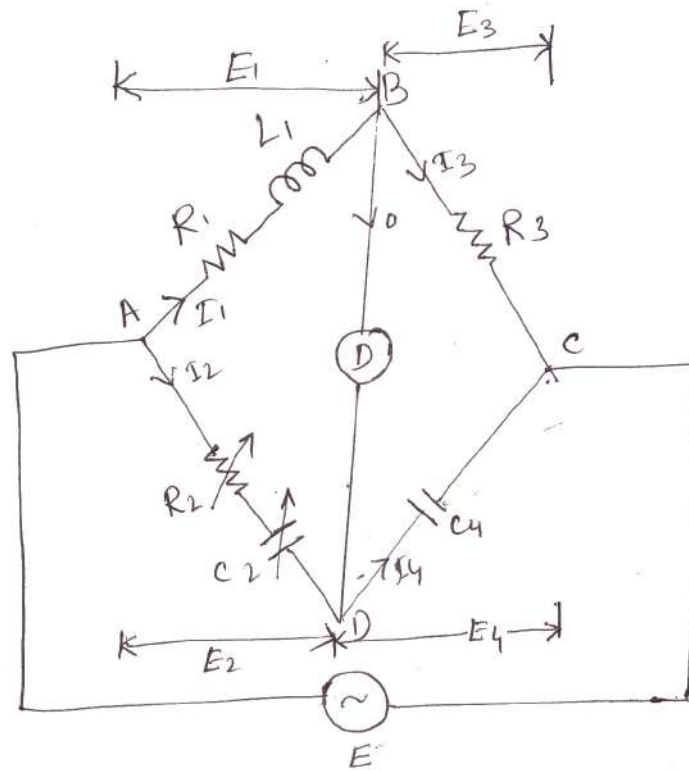


Fig 2.8 Owen's bridge

- $E_1 = I_1 R_1 + j I_1 X_1$
- I_4 leads E_4 by 90°

$$\dot{E} = \dot{E}_1 + \dot{E}_3$$

$$\dot{E}_2 = I_2 R_2 + \frac{I_2}{j\omega C_2}$$

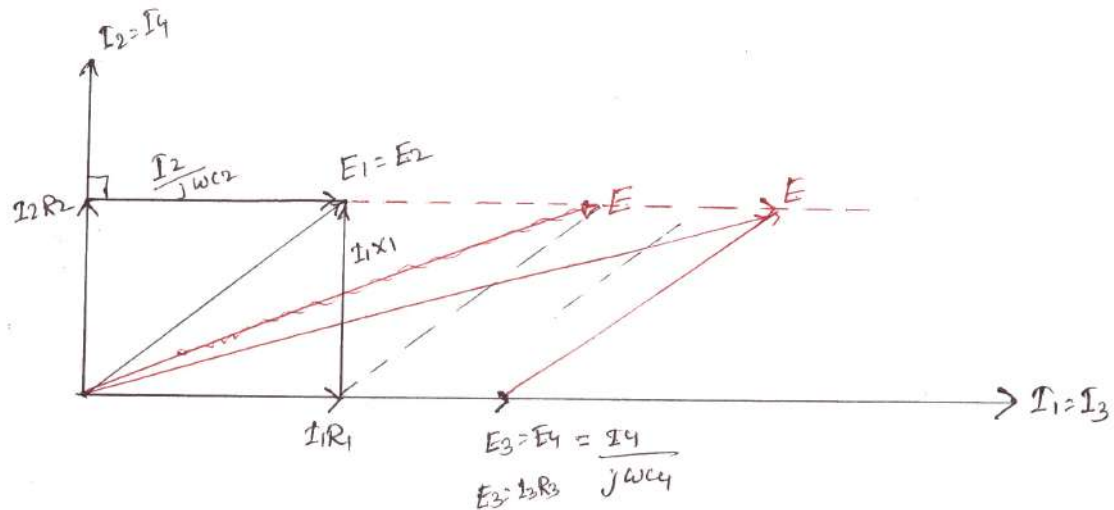


Fig 2.9 Phasor diagram of Owen's bridge

Balance condition, $\dot{Z}_1 \dot{Z}_4 = \dot{Z}_2 \dot{Z}_3$

$$Z_2 = R_2 + \frac{1}{j\omega C_2} = \frac{j\omega C_2 R_2 + 1}{j\omega C_2}$$

$$\therefore (R_1 + j\omega L_1) \times \frac{1}{j\omega C_4} = \frac{(1 + j\omega R_2 C_2) \times R_3}{j\omega C_2}$$

$$C_2 (R_1 + j\omega L_1) = R_3 C_4 (1 + j\omega R_2 C_2)$$

$$R_1 C_2 + j\omega L_1 C_2 = R_3 C_4 + j\omega R_2 C_2 R_3 C_4$$

Comparing real terms,

$$R_1 C_2 = R_3 C_4$$

$$R_1 = \frac{R_3 C_4}{C_2}$$

Comparing imaginary terms,

$$\omega L_1 C_2 = \omega R_2 C_2 R_3 C_4$$

$$L_1 = R_2 R_3 C_4$$

$$Q\text{-factor} = \frac{\omega L_1}{R_1} = \frac{\omega R_2 R_3 C_4 C_2}{R_3 C_4}$$

$$Q = \omega R_2 C_2$$

Advantages

- ✓ Expression for R_1 and L_1 are simple.
- ✓ R_1 and L_1 are independent of Frequency.

Disadvantages

- ✓ The Circuits used two capacitors.
- ✓ Variable capacitor is costly.
- ✓ Q-factor range is restricted.